FitBuddy: Project Workflow Guide

AI Fitness Plan Generator using Gemini Models | End-to-End Workflow Document

1. Overview of Project Workflow

The ***FitBuddy*** project workflow outlines the step-by-step lifecycle of user interaction, data processing, generative AI execution, and response rendering within the FastAPI application framework.

> **Workflow Objective:** Map out the clear sequence of operations from user input submission on the frontend to Gemini model inference and final interactive display on the browser dashboard.

2. Detailed Workflow Steps

- **Step 1: User Data Input**
  - The user fills out personal attributes (age, weight, height, primary goal, dietary preferences) on the HTML frontend form.
- **Step 2: Backend Request Routing**
  - The form data is transmitted via an HTTP POST request to the FastAPI endpoint (*/generate*) managed inside *routes.py*.
- **Step 3: Prompt Engineering & Gemini Processing**
  - The backend constructs a highly specialized prompt incorporating the user's parameters and invokes the Google Gemini Pro model (*gemini-pro*) via the SDK.
- **Step 4: Template Rendering & Response**
  - The generated markdown workout and diet regimen is parsed and passed through Jinja2 templates to render a responsive visual dashboard for the user.

3. Workflow Architecture Diagram Summary

User Browser (HTML Form) $ ightarrow$ FastAPI Router (*routes.py*) $ ightarrow$ Gemini Pro AI Engine $ ightarrow$ Jinja2 Template Engine $ ightarrow$ Rendered Dashboard

4. Summary

This streamlined workflow ensures low latency, high accuracy, and a seamless user experience throughout the ***FitBuddy*** application lifecycle.